Programming Fundamentals

LAB-02 Submission

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Program: **Computer Science**

Section: **(E)**

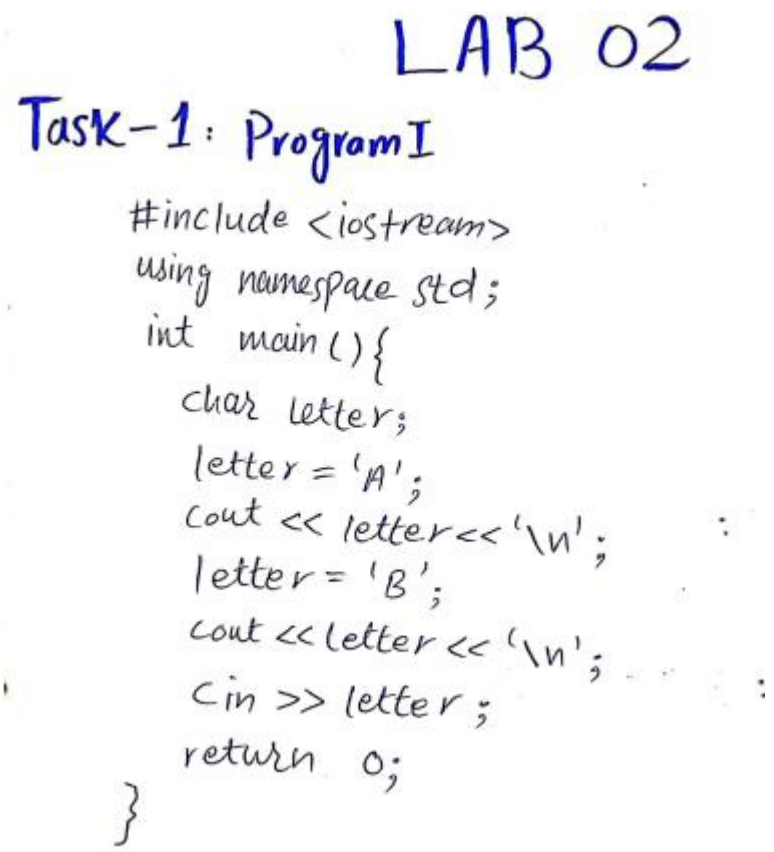
Submitted to: **Sir Asad Majeed**

Date: **11/10/2026**

LAB Task-01: Program-I

Question: Run the following program and check the output.

Handwritten Code:



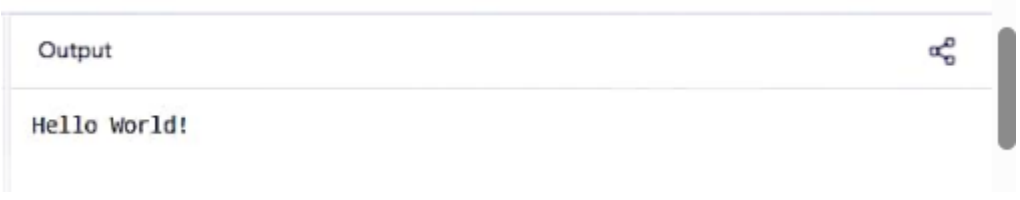
Laptop Run Code:

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;

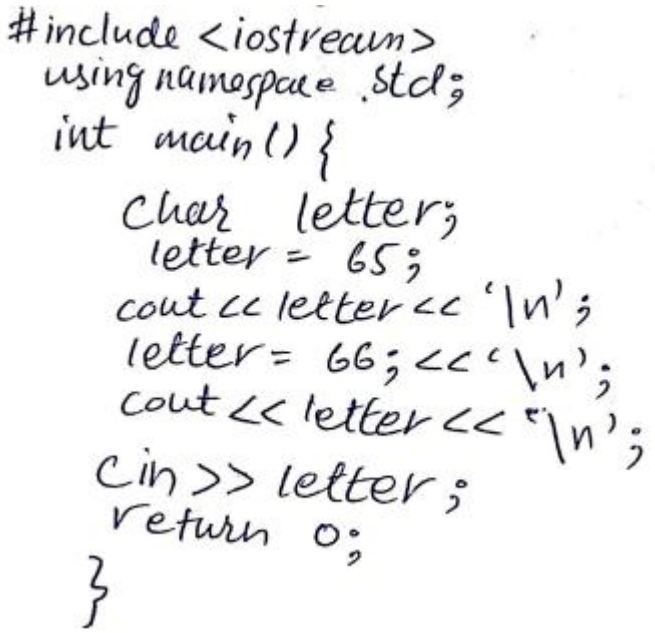
    return 0;
}
```

OUTPUT



LAB Task-01: Program-II

Handwritten code:



```
#include <iostream>
using namespace std;
int main() {
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66; << '\n';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Laptop Run Code:

```
#include <iostream>
using namespace std;

int main()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

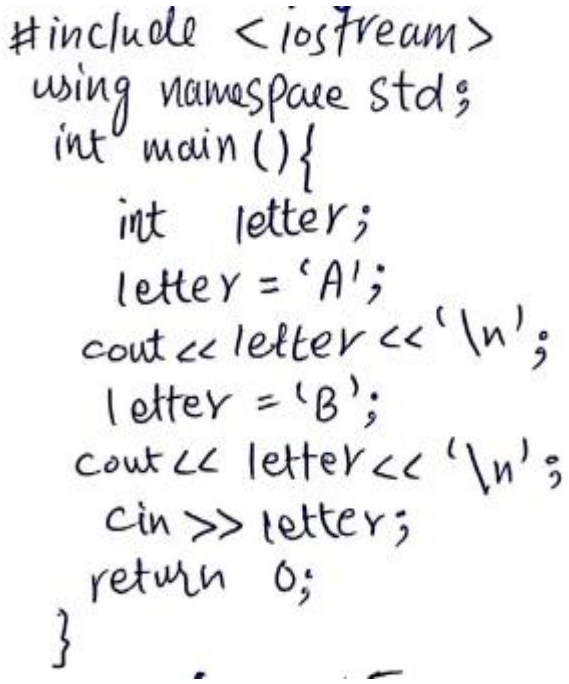
OUTPUT:

```
Output

A
B
```

LAB Task-01: Program-III

Handwritten Code:



```
#include <iostream>
using namespace std;
int main(){
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Laptop Run Code:

```
#include <iostream>
using namespace std;

int main()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

OUTPUT:

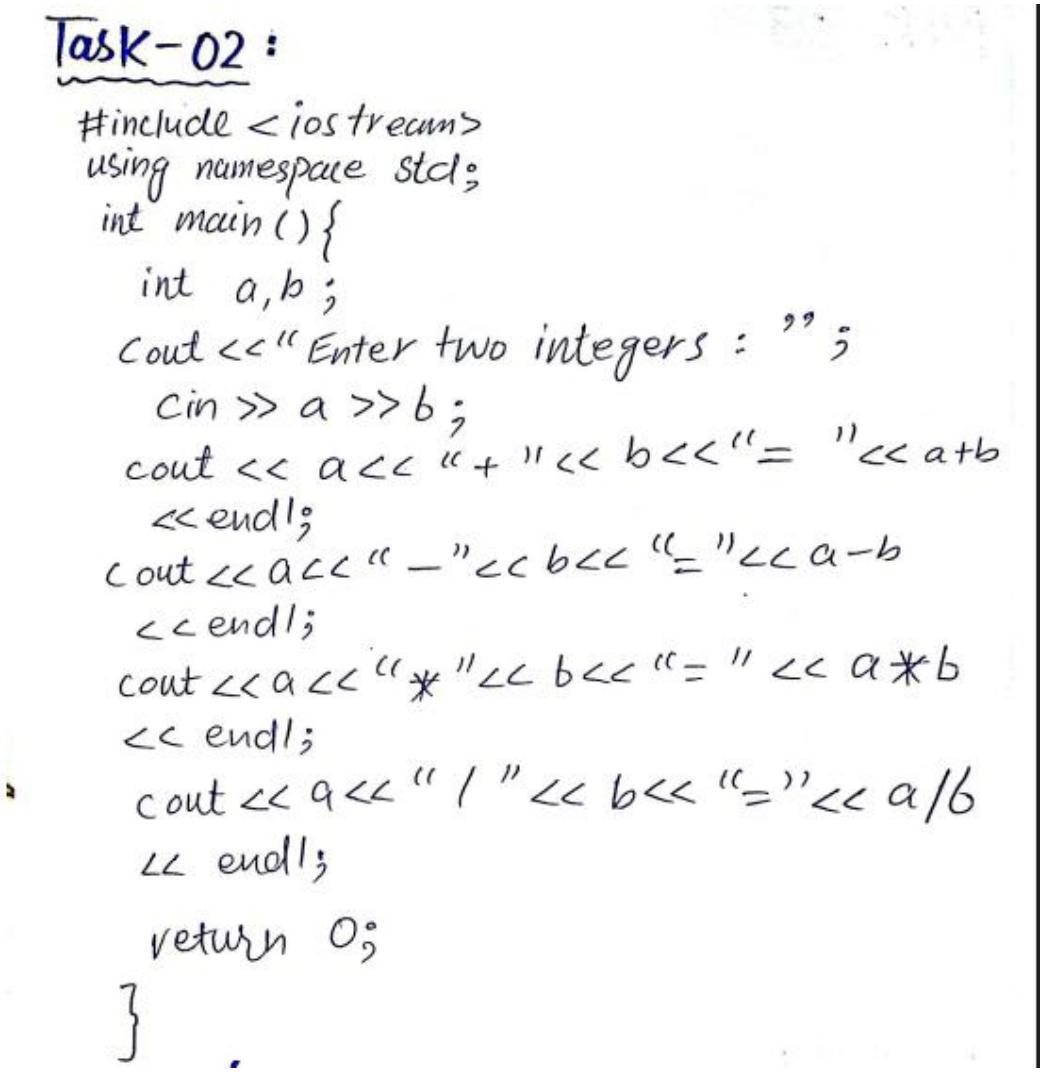
```
Output

65
66
```

LAB Task-02

Question: Take two integers as input from the user apply athematic operations on them (+,-,*, /) as print them on screen.

Handwritten Code:



Task-02 :

```
#include <iostream>
using namespace std;
int main(){
  int a,b;
 cout<<"Enter two integers : ";
  cin >> a >>b;
 cout << a<< " + " << b<<"= "<< a+b
  <<endl;
cout<<a<<" - "<<b<<"= "<<a-b
 <<endl;
cout<<a<<"* "<< b<<"= " << a*b
<< endl;
 cout<< a<<" / "<< b<<"="<< a/b
 << endl;
 return 0;
}
```

Laptop Run Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int a,b;
7      cout << "Enter two integers: ";
8      cin >> a >> b;
9
10     cout << a << " + " << b << " = " << a + b << endl;
11     cout << a << " - " << b << " = " << a - b << endl;
12     cout << a << " * " << b << " = " << a * b << endl;
13     cout << a << " / " << b << " = " << a / b << endl;
14     return 0;
15 }
```

OUTPUT:

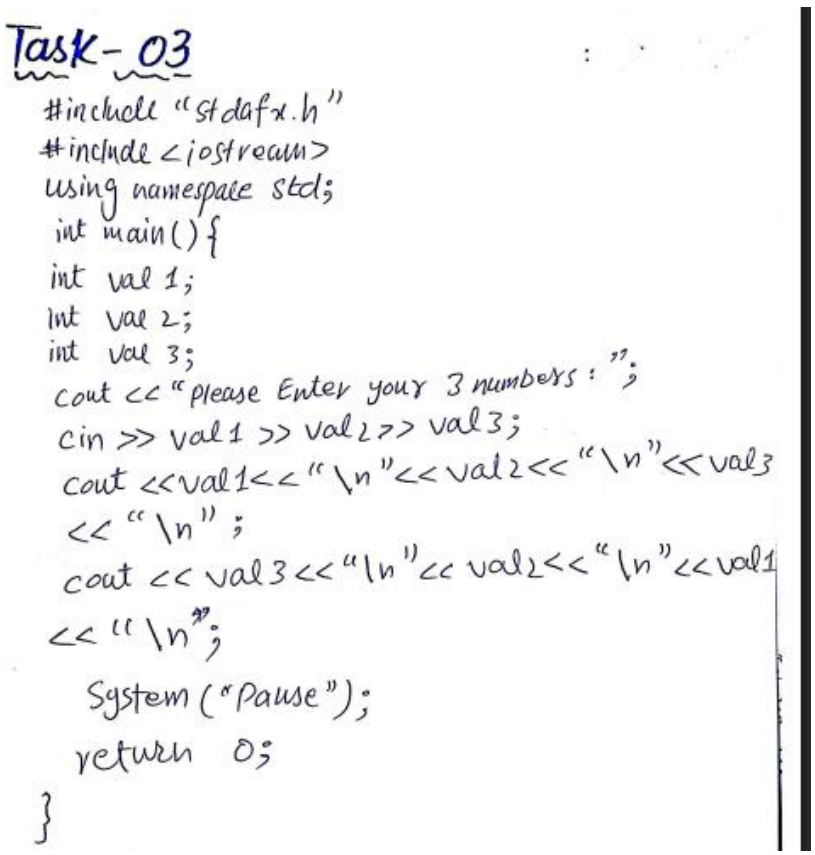
```
Output

Enter two integers: 10 2
10 + 2 = 12
10 - 2 = 8
10 * 2 = 20
10 / 2 = 5
```

LAB Task-03

Question: Write Output of the following Code.

Handwritten Code:



Task-03

```
#include "stdafx.h"
#include <iostream>
using namespace std;
int main(){
int val1;
int val2;
int val3;
cout << "please Enter your 3 numbers : ";
cin >> val1 >> val2 >> val3;
cout <<val1<< "\n"<< val2<< "\n"<<val3
<< "\n" ;
cout << val3<< "\n"<< val2<< "\n"<<val1
<< "\n";
  System ("Pause");
  return 0;
}
```

Laptop run code:

```cpp
#include<iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;
    cout << "Please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
    return 0;
}
```

OUTPUT:

Output

```
Please enter your 3 numbers:2 5 7
2
5
7
7
5
2
```

LAB Task-04

Question: Write a C++ program to calculate the distance between two points.

Handwritten Code:

Task-4 : write a C++ program to calculate the distance b/w two Points.

Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Program:

```
#include <iostream>
#include <cmath>
using namespace std;
int main () {
  double x1, y1, x2, y2, distance;
  cout << "Enter x1 and y1 : ";
  cin >> x1 >> y1 ;
  cout << " Enter x2 and y2 : ";
  cin >> x2 >> y2 ;
  distance = sqrt(pow(x2 - x1, 2)+pow(y2 - y1,2));
  cout << "distance = " << distance << endl;
  return 0;
}
```

LAPTOP RUN CODE:

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2, distance;

    cout << "Enter x1 and y1: ";
    cin >> x1 >> y1;
    cout << "Enter x2 and y2: ";
    cin >> x2 >> y2;

    distance = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

    cout << "Distance = " << distance << endl;

    return 0;
}
```

OUTPUT:

Output

```
Enter x1 and y1: 1 2
Enter x2 and y2: 4 6
Distance = 5
```